

CRAFT Prompt Generation Framework for Teachers

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Abstract

Generative artificial intelligence (GenAI) tools such as ChatGPT, Gemini, and Claude are transforming instructional design, yet most PK–12 educators lack effective strategies for engaging with them. To address this gap, we developed the CRAFT framework—a teacher-centered model for GenAI prompt design that emphasizes Context, Role, Audience, Format, and Tone. CRAFT translates principles from prompt engineering, instructional design, and Universal Design for Learning into a practical structure aligned with teachers’ professional planning language. We implemented the framework in a professional development program with 94 PK–6 teachers who used GenAI to create lessons integrating computational thinking. Their 688 prompts and AI-generated responses were coded across 30 analytical features representing critique, launch behaviors, refinements, and supplemental material generation. Findings suggest that CRAFT helped teachers produce standards-aligned, differentiated, and instructionally coherent lessons while increasing confidence, creativity, and reflective engagement with GenAI tools.

Keywords

Generative AI, Prompt Engineering, CRAFT Framework, Computational Thinking, PK-12 Lesson Planning

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1 Introduction

Generative artificial intelligence (GenAI) tools such as ChatGPT, Gemini, and Claude are transforming education by enabling teachers to design, adapt, and personalize instructional materials. However, most PK-12 educators lack clear strategies for effectively engaging with these tools. To address this gap, we developed the CRAFT framework—a teacher-centered model built around five key prompt elements: Context, Role, Audience, Format, and Tone. Unlike traditional prompt-engineering approaches requiring technical expertise, CRAFT aligns with teachers’ professional language and instructional design practices. To evaluate its effectiveness, we implemented CRAFT in a professional development (PD) program with 94 PK–6 teachers, collecting their AI prompts and the corresponding generative outputs. We analyzed these teacher–AI interactions using a 30-feature coding framework capturing critique, launch behaviors, procedural actions, refinements, and responses, allowing for a detailed examination of how educators iteratively design, evaluate, and refine lessons in collaboration with GenAI tools.

Through this study, we aim to contribute to the growing body of research on teacher–AI collaboration by (a) introducing the CRAFT framework as a novel, accessible tool for prompt design in PK–12 education; (b) analyzing how teachers operationalize and adapt CRAFT principles during authentic lesson development; and (c) identifying patterns of critique, refinement, and supplemental resource generation that characterize productive teacher–AI interaction. Our findings highlight the potential of CRAFT to not only improve the quality of AI-generated instructional materials but also to promote teacher agency, creativity, and reflective practice in the age of generative AI.

2 CRAFT Prompting Framework

The **CRAFT (Context, Role, Audience, Format, and Tone)** prompting framework is designed for educators—particularly those without prior GenAI experience or technical training—to leverage large language models such as ChatGPT to support their instructional goals. **Context** defines the instructional setting, including grade level, subject area, and topic (e.g., “Create a 4th grade science lesson on the water cycle”). **Role** encourages the teacher to assign the GenAI a professional identity aligned with educational expertise (e.g., “You are an experienced elementary science teacher”), which helps the AI produce more pedagogically appropriate and relatable content. **Audience** invites teachers to specify characteristics of the learners for whom the lesson is intended, allowing for differentiation based on student needs, language proficiency, or prior knowledge. **Format** directs the AI to output materials in a structure that is immediately usable in classrooms, such as a lesson plan, worksheet, or discussion guide. Finally, **Tone** ensures the instructional voice is developmentally appropriate, engaging, and aligned with the teacher’s classroom culture (e.g., “Use a curious and encouraging tone”). [Figure in poster]

The CRAFT framework is informed by three complementary fields in education: 1) prompt engineering [4], 2) instructional design, particularly the ADDIE model [1], and 3) Universal Design for Learning (UDL) [2]. The CRAFT framework is also familiar to teachers who utilize the RAFT [Role, Audience, Format, and Topic] strategy [3], which is most commonly used to teach writing.

3 Implementing CRAFT

In this study, we examine how 94 PK-6 teachers utilized GenAI in order to develop lesson plans and associated resources for integrating computational thinking (CT) into content instruction. Teachers came from 24 primarily rural districts across one Southeastern state. Teachers had an average of 12.2 years of experience and worked in a variety of P-8 settings, including classrooms, STEM enrichment, and as instructional coaches. Half of the teachers reported no prior experience using GenAI for instructional planning. After learning about CT integration during a week-long PD, teachers first developed their prompt engineering skills through guided practice and the use of the CRAFT prompting framework and subsequently advanced to a sequence of activities aimed at cultivating their ability to critique, refine, and adapt AI-generated lessons.

A sample CRAFT-based prompt used by a teacher to generate a lesson plan is as follows:

“You are an experienced 3rd grade music teacher. Create a 45-minute lesson plan on the so-fa syllables (Do, Re, Mi, Fa, So, La, Ti, Do). The lesson should include: Lesson Content (What are the phases or parts of the lesson? e.g., Hook, Read Aloud, Discussion, Activity, Sharing, Wrap-Up. How long will each part take? What will teachers need to say or do? Do you have any accommodations or alternative ideas that teachers could use instead?) Book included and purpose (Which book are you using? Give title, author name, and year published. Why was this book a good match for this lesson?) Materials (List all physical or digital materials,...) Vocabulary (What new vocabulary is being introduced ...) Lesson Title (What will be a catchy, fun title ...) • “Step by Step with Do-Re-Mi!” These are the standards: Computer Science Standards • Describe a daily task as a sequence of steps. • Use an ordered list of steps (sequential execution) and simple control structures to complete a task. Content Standards (Music) • Listen to, analyze, and describe music. • Perform music using correct pitch and rhythm. • Connect music learning to personal experience.”

4 Assessing CRAFT

In order to analyze the types of launches teachers used and their reflections on their process, teachers’ prompts and reflections on the use of GenAI were qualitatively coded into 30 categories by at least two members of the research team. Additional details on the coding scheme and feature generation will be included in the poster. Overall, the 94 teachers used 688 prompts, an average of 7.43 prompts per teacher. The minimum number of prompts used was 1, and the maximum was 26. Teachers felt that CRAFT helped them to create lessons that addressed content and CS standards, were differentiated for the needs of their students, and included a variety of materials, including assessments, teacher materials, and student materials. As one teacher said, “I was amazed at how well it went! The CRAFT method really helped me convey the details that made it generate something very close to what I envisioned.”

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